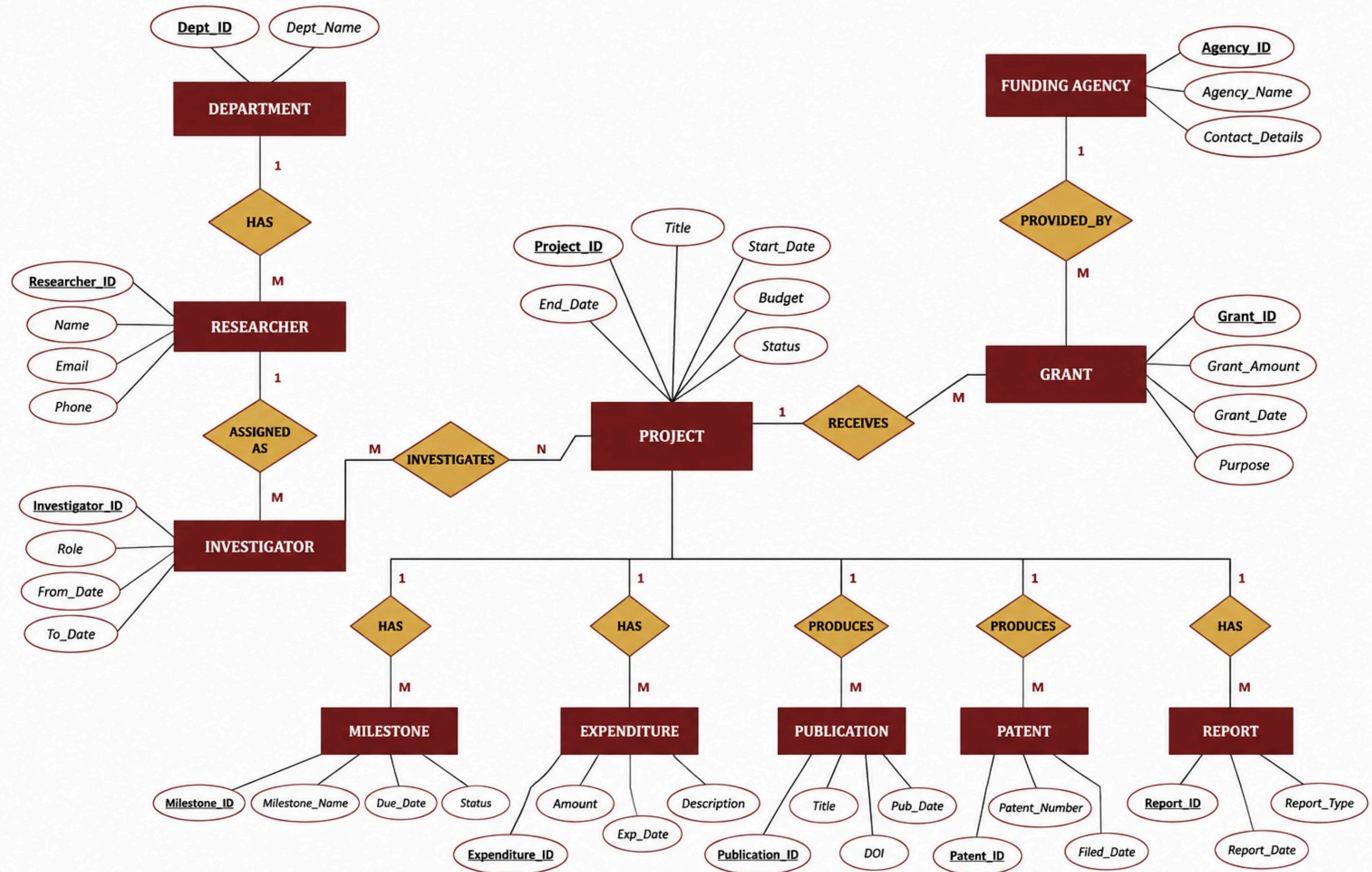


DBMS Assignment 1

Design and Implementation of a database management system for
Research Project, Publication and Grant Management System

Conceptual ER Diagram



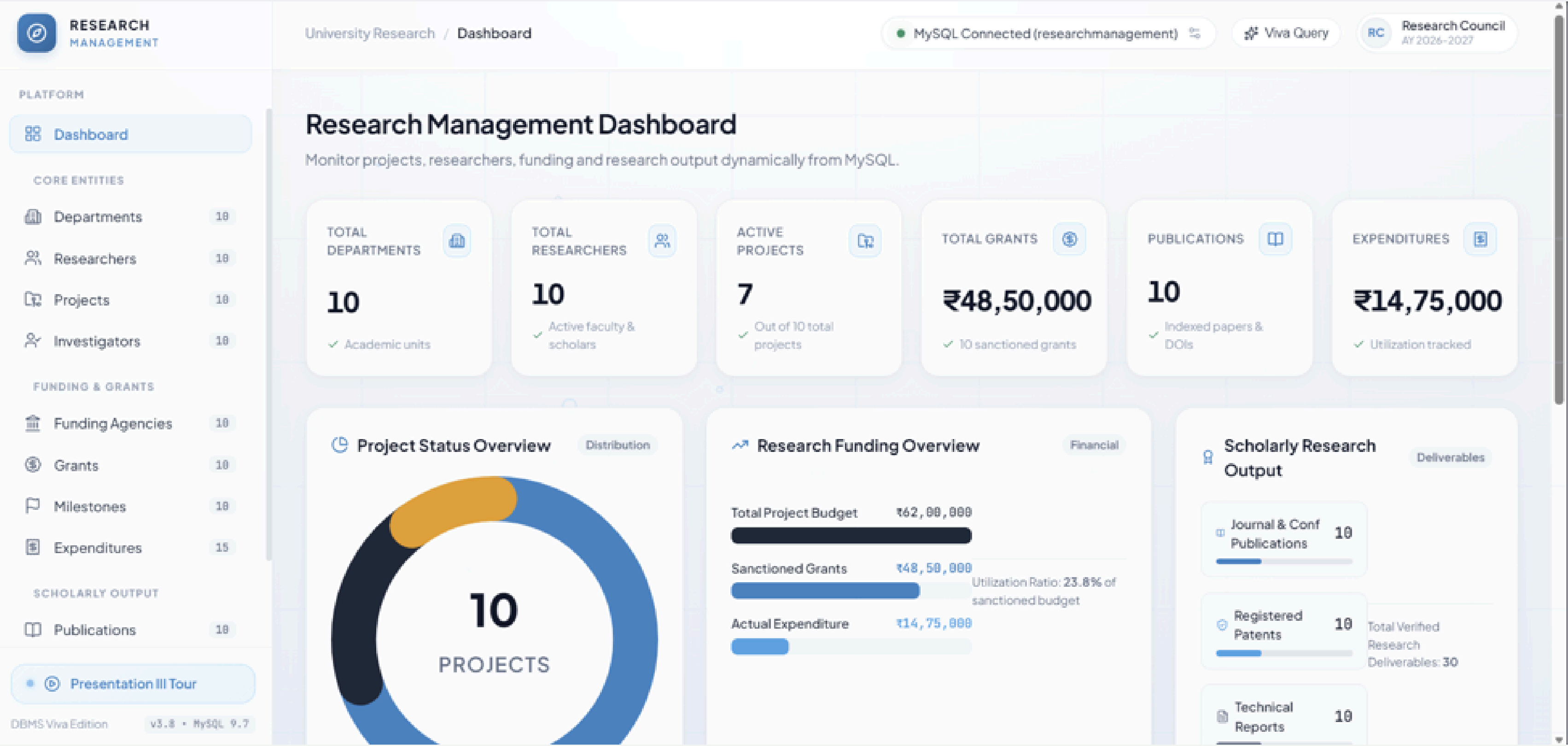
Relational Schema

RELATION	ATTRIBUTES
DEPARTMENT	Dept_ID (PK), Dept_Name
RESEARCHER	Researcher_ID (PK), Dept_ID (FK), Name, Email, Phone
PROJECT	Project_ID (PK), Title, Start_Date, End_Date, Budget, Status
INVESTIGATOR	Investigator_ID (PK), Researcher_ID (FK), Project_ID (FK), Role, From_Date, To_Date
FUNDING AGENCY	Agency_ID (PK), Agency_Name, Contact_Details
RESEARCH GRANT	Grant_ID (PK), Project_ID (FK), Agency_ID (FK), Grant_Amount, Grant_Date, Purpose
MILESTONE	Milestone_ID (PK), Project_ID (FK), Milestone_Name, Due_Date, Status
EXPENDITURE	Expenditure_ID (PK), Project_ID (FK), Amount, Exp_Date, Description
PUBLICATION	Publication_ID (PK), Project_ID (FK), Title, Pub_Date, DOI
PATENT	Patent_ID (PK), Project_ID (FK), Patent_Number, Filed_Date
REPORT	Report_ID (PK), Project_ID (FK), Report_Type, Report_Date

Main Entities

Entity	Purpose
Department	Stores academic/research departments
Researcher	Stores researcher information
Project	Stores research project details
Investigator	Represents researcher involvement in projects
Funding Agency	Stores organizations providing funding
Grant	Stores project funding details
Milestone	Tracks project milestones
Expenditure	Records project expenses
Publication	Stores research publications
Patent	Stores project-related patents
Report	Stores project/research reports

UI Dashboard



Screenshot of Researchers

RESEARCH MANAGEMENT

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Presentation III Tour

DBMS Viva Edition

v3.8 • MySQL 9.7

University Research / Researchers

MySQL Connected (researchmanagement) Viva Query RC Research Council AY 2026-2027

Export CSV Add Record

Researchers

Live MySQL records in researchmanagement.researcher

Search researcher records...

Total Records: 10 rows

RESEARCHER ID PK	DEPT ID FK	NAME	EMAIL	PHONE	ACTIONS
#181	#1	Aarav Sharma	aarav@university.edu	9876543210	
#182	#1	Priya Rao	priya@university.edu	9876543211	
#183	#2	Rahul Mehta	rahul@university.edu	9876543212	
#184	#3	Ananya Iyer	ananya@university.edu	9876543213	
#185	#5	Vikram Singh	vikram@university.edu	9876543214	
#186	#1	Madhu	madhu@university.edu	9876543217	
#187	#7	Arjun Reddy	arjun@university.edu	9876543216	
#188	#8	Sneha Patel	sneha@university.edu	9876543215	
#189	#9	Karan Malhotra	karang@university.edu	9876543218	
#110	#10	Meera Nair	meera@university.edu	9876543219	

Screenshot of Patents

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DBMS Viva Editionv3.8 • MySQL 9.7

University Research / Patents

MySQL Connected (researchmanagement)Viva QueryResearch CouncilAY 2026-2027

Export CSVAdd Record

Patents

Live MySQL records in researchmanagement.patent

Search patent records...

Total Records: 10 rows

PATENT ID PK	PROJECT ID FK	PATENT NUMBER	FILED DATE	ACTIONS
#901	#201	PAT-AI-001	1 Jun 2026	
#902	#202	PAT-AGR-002	10 Aug 2026	
#903	#203	PAT-ENE-003	5 Sept 2026	
#904	#204	PAT-AUTO-004	12 Jul 2026	
#905	#205	PAT-NLP-005	12 Sept 2026	
#906	#206	PAT-DIS-006	1 Oct 2026	
#907	#207	PAT-BIO-007	10 Oct 2026	
#908	#208	PAT-CLI-008	1 Nov 2026	
#909	#209	PAT-CYB-009	5 Oct 2026	
#910	#210	PAT-ROB-010	15 Nov 2026	

Screenshot of Projects

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University Research / Projects

MySQL Connected (researchmanagement)Viva QueryResearch CouncilAY 2026-2027

Projects

Live MySQL records in researchmanagement.project

Export CSVAdd Record

Search project records...Total Records: 10 rows

PROJECT ID PK	TITLE	START DATE	END DATE	BUDGET	STATUS	ACTIONS
#201	AI Based Healthcare Analysis	10 Jan 2026	20 Dec 2026	₹5,00,000	Active	
#202	Smart Agriculture System	15 Feb 2026	30 Oct 2026	₹3,00,000	Active	
#203	Sustainable Energy Optimization	1 Mar 2026	30 Nov 2026	₹7,50,000	Completed	
#204	Autonomous Vehicle Monitoring	20 Jan 2026	30 Sept 2026	₹4,50,000	Completed	
#205	Natural Language Processing for Education	1 Apr 2026	31 Mar 2027	₹9,00,000	Active	
#206	AI Powered Disease Detection	1 May 2026	30 Apr 2027	₹6,50,000	Active	
#207	Biomedical Signal Processing	1 Jun 2026	31 May 2027	₹5,50,000	Active	
#208	Climate Change Prediction Model	1 Jul 2026	30 Jun 2027	₹7,00,000	Active	
#209	Cybersecurity Threat Detection	15 May 2026	31 Mar 2027	₹6,00,000	Active	
#210	Robotic Assistance System	1 Aug 2026	31 Jul 2027	₹8,00,000	Pending	



Thank You